

Selected Agent :- Smart Exam Checker Agent
Problem in the context of agent :-

In the education system, Teachers manually check exam papers which takes a lot of time and effort. There is also a chance of human error during checking. To solve this problem, an intelligent agent can be developed that automatically checks students MCQ's papers, assign marks and generate results. The agent must ensure 100% accurate checking with minimum time.

Problem Formulation :-

a) Textual :-

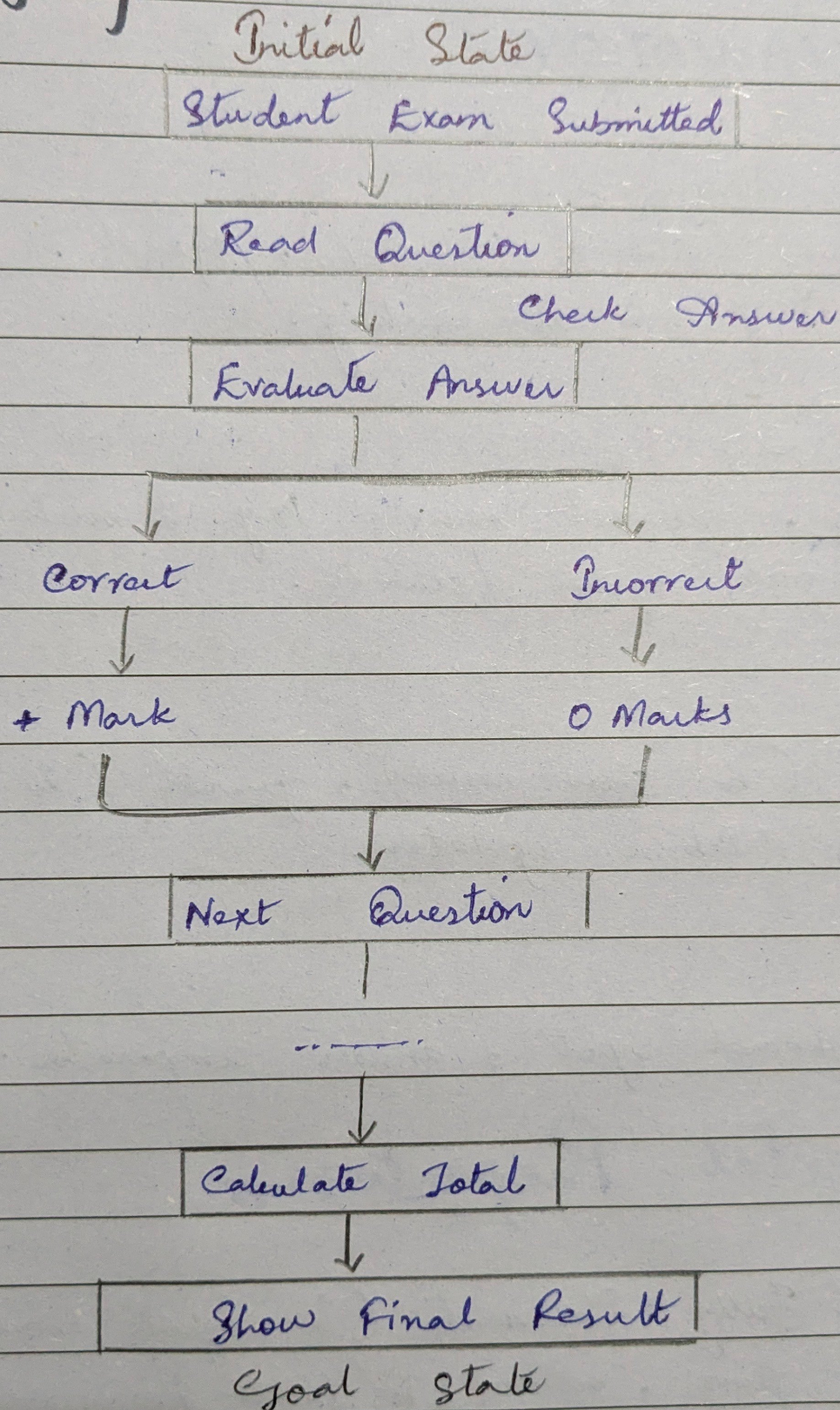
→ States :- Questions, student answers, correct answers, marks obtained, total marks, checked / unchecked status of each question.

→ Initial State :- All questions are unchecked, marks=0, student has submitted the answer sheet.

- Actions :- Read next answer, compare with correct answer, mark as correct / incorrect, add marks, display result.
- Transition Model :- When the agent checks one answer, the question becomes "checked". If the answer is correct, marks are added. If incorrect, no marks are added. After each question, total marks are updated.
- Goal Test :- All questions are checked, marks are calculated and final result is generated.
- Path Cost :- Time taken to check all papers + accuracy of checking. Goal is to complete checking with maximum accuracy in minimum time.

b) Graphical:-

Date _____



PEAS EVALUATION:-

→ Performance Measure :-

Checking accuracy (should be 100%), speed, zero human error, fairness in marking.

→ Environment :-

Answer sheets, correct answer key, marking scheme, computer system / scanner.

→ Actuators :-

Green display (to show results), printer (to print marksheets), database updater.

→ Sensors :-

Scanner, keyboard input, answer comparator.

Environmental Properties :-

Observability : Fully Observable (the agent know everything, nothing is hidden).

Deterministic vs Stochastic : Deterministic (every action has a clear result).

Episodic vs Sequential : Episodic (checking of one paper does not affect checking another paper).

Static vs Dynamic :- Static (Environment remains same while paper is being checked).

Discrete vs Continuous :- ~~States~~ are Discrete (States are clean - checked / unchecked, correct / incorrect).

Single vs Multi-Agent :- Single Agent (Only one agent is checking the papers).